

Space Agencies Report Increased Risk of Space Rock Striking Earth in 2032

Scientists say the threat of a large space rock striking the Earth in 2032 has increased.

Recent calculations suggest there is a small chance - perhaps around 1 to 3 percent - the space rock 2024 YR4 will hit Earth in 2032. This also means there is around a 97 to 99 percent chance it will safely pass our planet in that year. The odds of a strike will likely continue to go up and down as the asteroid's path around the sun is better understood.

And scientists say there is a good chance the risk will drop to zero.

NASA and the European Space Agency's Webb Space Telescope will observe this near-Earth asteroid in March before the object becomes impossible to see. Once that happens, scientists will have to wait until 2028 when it passes our way again.

What is an asteroid?

Asteroids are space rocks orbiting the sun.

Asteroids are much smaller than planets. Scientists believe they are the leftovers from the solar system's formation 4.6 billion years ago.

There are millions of asteroids orbiting the sun between Mars and Jupiter. That area is known as the main asteroid belt. The asteroids sometimes get pushed out of the belt and can end up elsewhere.

How do scientists follow worrisome asteroids?

A telescope in Chile discovered the asteroid 2024 YR4 in late December. It is estimated to measure between 40 and 90 meters across. NASA said observations by the Webb space telescope should provide a more exact measurement.

At first, NASA and the European Space Agency put the odds of a strike at just over one percent. But later calculations suggested the odds had risen to 2 or even 3 percent. NASA described the likelihood of a strike as "extremely low."

Until scientists have a better understanding of the asteroid's path around the sun, they warned that the odds will continue to change — and quite possibly fall to zero.

"You don't have to be worried about anything. It's a curiosity," said Larry Denneau. He is a software engineer at the University of Hawaii working with ATLAS, a program to identify

asteroids that might endanger Earth. That effort first spotted the asteroid. Denneau said, “Don’t panic. Let the process play out, and we’ll have a for-sure answer.”

In 2021, NASA said another possibly worrisome asteroid, Apophis, did not present a risk to Earth. NASA made the announcement after additional telescope observations ruled out any chance of it hitting Earth in 2068.

Should we worry about asteroid 2024 YR4?

Experts say it is too soon to worry about 2024 YR4.

Paul Chodas is director of NASA’s Center for Near-Earth Object Studies. Chodas said in an email to the Associated Press, “No one should be concerned that the impact probability is rising. This is the behavior our team expected.”

Chodas added, “To be clear, we expect the impact probability to drop to zero at some point.”

Since the asteroid’s size and orbit are uncertain, it is unclear where it might hit and what the possible effects would be should it strike Earth. ESA said, if the asteroid is smaller, the effect might be similar to the Tunguska event that flattened thousands of square kilometers of forest in Siberia in 1908. But if the asteroid is larger, close to 100 meters, the effects “would be significantly worse.”

Chodas said once the Webb telescope measures the asteroid’s size, NASA can predict “how serious an impact this asteroid could produce and how difficult a task it might be to deflect this asteroid.”

NASA already has some experience moving an asteroid. The space agency’s Dart spacecraft hit a harmless asteroid in 2022. This was the first planetary defense test of its kind, changing the asteroid’s orbit around its larger companion asteroid.

I’m John Russell.

Marcia Dunn reported this story for the Associated Press. John Russell adapted it for VOA Learning English.

Vocabulary

odds - n. (พหุพจน์) โอกาสที่สิ่งใดสิ่งหนึ่งจะเกิดขึ้น

curiosity - n. ความปรารถนาอย่างแรงกล้าที่จะเรียนรู้เกี่ยวกับบางสิ่ง

panic - v. ตื่นตระหนก, ถูกครอบงำด้วยความกลัวอย่างรุนแรง

probability - n. โอกาสที่เหตุการณ์จะเกิดขึ้น

impact - n. การชน, เหตุการณ์ที่วัตถุหนึ่งพุ่งชนหรือกระแทกเข้ากับอีกวัตถุหนึ่ง

significantly - adv. อย่างมีนัยสำคัญ, อย่างมาก

deflect - v. เบี่ยงเบน, ทำให้วัตถุเคลื่อนที่ไปในทิศทางอื่นโดยอาจชนและเปลี่ยนเส้นทางเล็กน้อย
companion - n. วัตถุที่เชื่อมโยงกับวัตถุอื่นอย่างใกล้ชิด มักจะด้วยแรงโน้มถ่วง

Reading Comprehension Quiz

1. What is the main subject of concern mentioned in the initial paragraphs regarding the year 2032?

- A. The discovery of a new comet.
- B. The increasing threat of a large space rock striking Earth.
- C. The development of new space telescopes.
- D. The decrease in solar activity.

2. What is the current estimated probability range for asteroid 2024 YR4 hitting Earth in 2032?

- A. 97 to 99 percent.
- B. Less than 1 percent.
- C. Around 1 to 3 percent.
- D. Exactly 50 percent.

3. According to scientists, what is a potential future outcome regarding the risk of an impact?

- A. The risk is guaranteed to increase.
- B. The risk will definitely remain the same.
- C. There's a good chance the risk will drop to zero.
- D. The risk is impossible to predict.

4. When will the Webb Space Telescope observe the near-Earth asteroid before it becomes impossible to see?

- A. 2032
- B. 2028
- C. March
- D. Late December

5. Based on the article, what is the fundamental definition of an asteroid?

- A. A planet-sized object.
- B. A space rock orbiting the sun.
- C. A moon of another planet.

D. A comet from outside the solar system.

6. What do scientists believe asteroids are in relation to the solar system's history?

- A. Newly formed celestial bodies.
- B. Remnants from Earth's early history.
- C. Leftovers from the solar system's formation 4.6 billion years ago.
- D. Pieces of broken planets.

7. Where is the main asteroid belt typically located?

- A. Between Earth and Mars.
- B. Beyond Jupiter.
- C. Between Mars and Jupiter.
- D. Close to the sun.

8. How large is asteroid 2024 YR4 initially estimated to be?

- A. Over 100 meters.
- B. Exactly 40 meters.
- C. Between 40 and 90 meters across.
- D. Less than 10 meters.

9. What was the initial estimated probability of a strike according to NASA and ESA?

- A. 2 or 3 percent.
- B. Just over one percent.
- C. Extremely low.
- D. Zero percent.

10. What is required for scientists to make more definitive predictions about the asteroid's path?

- A. More funding for new telescopes.
- B. A better understanding of the asteroid's path around the sun.
- C. Collaboration with other space agencies.
- D. The asteroid to pass by Earth again.

11. What is Larry Denneau's advice regarding the asteroid 2024 YR4?

- A. To immediately prepare for impact.
- B. To panic and demand answers.
- C. Not to worry and let the process play out.

D. To ignore all scientific warnings.

12. What was the conclusion regarding asteroid Apophis in 2021?

- A. It posed a definite risk to Earth in 2068.
- B. It did not present a risk to Earth.
- C. Its path was still uncertain.
- D. It would be deflected by NASA.

13. According to Paul Chodas, why shouldn't people be concerned about the rising impact probability?

- A. The asteroid is too small to cause harm.
- B. It is the expected behavior of such calculations.
- C. NASA has already developed a deflection plan.
- D. The asteroid will burn up in the atmosphere.

14. What was the estimated effect of the Tunguska event mentioned in the text?

- A. It created a large crater.
- B. It caused a global climate change.
- C. It flattened thousands of square kilometers of forest.
- D. It led to a mass extinction.

15. What was the purpose of NASA's Dart spacecraft mission in 2022?

- A. To collect samples from an asteroid.
- B. To test planetary defense by changing an asteroid's orbit.
- C. To observe a distant asteroid.
- D. To search for new asteroids.